ASSESSMENT EXERCISE

Multiple Choice Questions

1. Why is transparency important in AI ethics?
(a) It makes AI systems smarter
(b) It ensures that AI algorithms are kept secret
(c) It helps users understand how AI systems make decisions
(d) It slows down the decision-making process

2. In the context of AI ethics, what is the problem of inclusion?

(a) Ensuring AI algorithms are exclusive
(b) Making AI systems treat everyone fairly and inclusively
(c) Excluding certain groups intentionally
(d) Ignoring ethical considerations in AI development

3. What is one ethical issue related to AI and privacy?

(a) Using AI to enhance personal privacy
(b) AI systems sharing personal information without permission
(c) Privacy has no relevance to AI
(d) AI is not capable of handling personal data

4. What is the Cambridge Analytica scandal related to?

(a) AI bias in hiring
(b) Privacy concerns with AI
(c) Job displacement due to automation
(d) Ethical issues in self-driving cars

5. What does AI bias refer to?

(a) The tendency of artificial intelligence to favor certain individuals or groups over others.
(b) Unfair or prejudiced outcomes resulting from biased data or algorithms.
(c) The accuracy of AI systems in recognizing diverse patterns.
(d) The speed at which AI systems process information.

6. Which of the following is not a way in which AI bias can occur?

(a) Training Data Bias
(b) Algorithmic Bias
(c) Feedback Loop Bias
(d) Real-time Decision Bias

Very Short Answer Type Questions

7. How do ethical AI frameworks address unfair bias?

Ans: Ethical AI frameworks address unfair bias by using fair and balanced training data, testing AI systems for bias, and correcting unfair decisions to ensure equal treatment for everyone.

8. How does ethical AI contribute to human-centered principles?

Ans: Ethical AI respects human rights, privacy, fairness, and safety. It ensures that AI systems benefit people, protect their interests, and do not discriminate against any group.

9. Name some ethical issues of AI.

Ans: Some ethical issues of AI are:

- AI bias and discrimination
- Privacy and data security
- Lack of transparency
- Lack of accountability
- Unequal access to AI technology
- Safety and potential harm to people

Short Answer Type Questions

10. What is AI bias? Give an example.

Ans: AI bias means unfair or prejudiced results produced by an AI system because of biased data, algorithms, or assumptions.

Example: If an AI recruitment system is trained on data that favors male candidates, it may unfairly reject qualified female candidates. This is an example of AI bias.

11. Discuss the bias in data collection.

Ans: Bias in data collection occurs when the data collected to train an AI model is incomplete, unbalanced, or unfairly represents certain groups.

For example, if a facial recognition system is trained mainly on images of people with light skin, it may not recognize people with dark skin accurately.

Bias in data collection can lead to unfair decisions and discrimination. Therefore, training data should be accurate, diverse, balanced, and representative of all groups.

Long Answer Type Question

12. What are the possible reasons of AI bias?

Ans: AI bias refers to unfair or prejudiced results produced by an AI system. It can occur due to several reasons:

1. Human Bias in Decisions

Human biases can enter AI systems during their development. Programmers may unintentionally include their own beliefs or prejudices in algorithms, leading to unfair results.

2. Flawed and Unbalanced Data Collection

If the data used to train an AI system is incomplete, incorrect, or biased, the system may learn those biases and produce unfair decisions.

Example: If a hiring system is trained mainly on records of male employees, it may favor male candidates over equally qualified female candidates.

3. Under- or Over-Representation of Specific Features

When certain groups, such as particular races or genders, are under-represented or over-represented in training data, AI may not perform equally well for everyone.

Example: A facial recognition system trained mostly on adults may have difficulty recognizing children's faces accurately.

4. Wrong Assumptions

If developers make incorrect assumptions about the data or the real world, an AI system may learn these assumptions and produce biased results.

Example: Assuming that people from a particular area are less trustworthy may lead to unfair decisions about them.

5. No Proper Bias Testing

If an AI system is not properly tested for bias before being used, unfair patterns may remain undetected and cause problems later.

6. No Bias Mitigation
Even after identifying bias, if developers do not take steps to reduce it, the AI system may continue producing unfair results.

AI bias can lead to discrimination and unfair decisions. To reduce it, developers should use diverse and balanced training data, test AI systems regularly, correct unfair patterns, and ensure that AI treats everyone fairly.